Discrete Random Variables & Binomial Distributions

Section 1: Expected Value of a Discrete Random Variable

1 A discrete random variable X has distribution: $P(X=0)=0.2, P(X=1)=0.5, P(X=2)=0.3$. Find $E(X)$.

a. 1.1

b. 1.0

c. 1.5

d. 0.8

2 A carnival game costs 5 dollars to play. There is a 10% chance to win 20 dollars, a 30% chance to win 5 dollars, and a 60% chance to win nothing. What is the player's expected net gain?

a. -1.50 dollars

b. 3.50 dollars

c. -0.50 dollars

d. 0.00 dollars

3 If $E(X)=12$ and $Y=3X-5$, what is $E(Y)$?

a. 31

b. 36

c. 41

d. 12

Section 2: Variance & Standard Deviation of Combinations

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4 Independent random variables X and Y have $\text{SD}(X) = 3$ and $\text{SD}(Y) = 4$. What is $\text{SD}(X - Y)$?

a. 5

b. 1

c. -1

d. 7

5 If $\text{SD}(X) = 6$, what is $\text{SD}(4X + 10)$?

a. 24

b. 34

c. 144

d. 6

6 Why must variances be added rather than standard deviations when combining independent variables?

a. Standard deviations are not in squared units, while the Pythagorean relationship of orthogonal independent variables governs variances

b. Standard deviations can be negative

c. Adding standard deviations always gives an imaginary number

d. Because variance is always smaller than standard deviation

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Section 3: The Binomial Setting & Probability Formula

7 Which of the following is NOT an essential requirement for a binomial setting (BINS)?

a. The sample size must exceed 30

b. Binary outcomes (Success or Failure) for each trial

c. Independent trials

d. Fixed number of trials n, and same probability of success p

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8 A fair 6-sided die is rolled 5 times. What is the probability of rolling exactly two 4s?

a. $\binom{5}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$

b. $\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$

c. $\binom{5}{2}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^2$

d. $\frac{2}{5}$

9 Evaluate $\binom{5}{2}$.

a. 10

b. 20

c. 5

d. 15

Section 4: Mean & Standard Deviation of Binomial Distributions

10 A multiple-choice test has 20 questions, each with 4 options. If a student guesses completely at random, what are the mean and standard deviation of correct answers?

a. $\mu = 5.0, \sigma = 1.94$

b. $\mu = 5.0, \sigma = 3.75$

c. $\mu = 4.0, \sigma = 2.00$

d. $\mu = 5.0, \sigma = 15.0$

11 When is a Normal approximation to a binomial distribution considered valid?

a. When $np \geq 10$ and $n(1-p) \geq 10$

b. When $n \geq 30$

c. When $p = 0.5$ regardless of n

d. When $np(1-p) \geq 30$

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12 In a geometric distribution where $p = 0.2$, what is the expected number of trials until the first success?

a. 5

b. 0.2

c. 4

d. 25

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